

Week 13 Progress Report

StellarOrion HypersonicEdition — HIAD Validation & Topology Optimization

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Agenda

What's New (Git Diff & Progress)

Validation Progress

Optimization Progress

Before / After Topology

Performance References

Optimization Performance

What's New — Latest Commit

5bcb777 feat(hiad): optimization pipeline with BO, dual-unit display, validation overlay 21 Sep 2026

17 files changed, +758 / -225 (repo: StellarOrion_HypersonicEdition, code root stellarorion_program_proc)

- ▶ Replaced MoP with **Bayesian Optimization** (GP Matern 5/2 + Expected Improvement, 70 evaluations)
- ▶ Optimization result: **Cd 1.607 → 1.455 (−9.45%)**
- ▶ Trajectory model altitude extended **50 km → 40 km**
- ▶ Added KM_TO_FT = 3280.84 dual-unit altitude display
- ▶ Added --validation CLI flag and _draw_validation_overlay()
- ▶ Added IRVE3_REFERENCE comparison in validation_pipeline.py
- ▶ Ada trajectory_output: **10 bug fixes** (Bugs #20–#29)
- ▶ Added Optimization_Attempt_1.md documentation

What's New — Files Touched in _proc

Core (commit 5bcbb77)

File	Δ
ada_pinn_wrapper.py	± 213
generate_outputs.py	± 220
validation_pipeline.py	± 97
hiad_optimizer.py	± 50
trajectory_output.adb	± 42
postprocess_validation.py	± 86
Optimization_Attempt_1.md	$+187$
hiad_optimization_results.json	± 28

Working tree now (uncommitted)

- ▶ New: optimization_comparison.png
- ▶ New: scripts/
render_optimization_comparison.py
- ▶ Mode-only churn: 14 text files (prove.sh $\times 2$, run_smooth.sh, plotting/validation/viewer tools, uiassets/main.ts) — 0-line content diffs
- ▶ Regenerated 20 UI plot PNGs (same byte sizes, re-rendered)

Recent history (progress): bb2e059 7 new Ada FFI wrappers + CLI args · f3b0051 dashboard Peak Heat Flux panel · 2716ed1 linearized trajectory, uniform VTU/MP4 · 3df86ad removed stale 0.55 m nose-radius references.

Validation Progress — Where We Are

- ▶ **29 Aug:** scalloped run truncated at 1000/2200 steps (Colima not started) → status **BLOCKED** (Validation Aug29_ValidationProgress.md)
- ▶ **1 Sep:** full 2200-step SPARTA DSMC completed for scalloped + smooth geometries
- ▶ **2 Sep:** post-processing complete — unified comparison table + convergence audit generated by `validation_pipeline.py`
- ▶ DSMC → **PINN hand-off** implemented: step 2200 → step 3×10^8 , trajectory 120 km → 50 km

Sim conditions (both geometries): alt 51.82 km, V 3379 m/s, Mach 10.29, 2200 steps, $\Delta t = 1 \mu s$, 6 MPI ranks, $\approx 1,387,500$ particles, 19,321 cells, 76 surf elements, $T_{wall} = 1000$ K.

Validation Progress — Scalloped vs Smooth vs IRVE-3

Metric	Scalloped	Smooth	Δ (scal.-smooth.)
Peak drag [N]	62,726	57,468	+9.1%
Avg. drag [N]	49,123	45,157	+8.8%
Mean C_d	1.5821	1.4544	+8.8%
Peak heat flux [W/cm ²]	428.80	424.91	+0.9%
SG stagnation [W/cm ²]	12.20*	—	vs IRVE-3 14.36
Mean heat load [J/cm ²]	165.72 [†]	—	vs IRVE-3 195.06
Peak G-load	16.83	—	vs target 19.7

*SG at hardcoded Rapisarda baseline ($R_n=0.55$ m, $V=2700$ m/s); at actual sim conditions true SG ≈ 25.1 W/cm².

[†]Single-point DSMC — not trajectory-integrated (IRVE-3 195.06 is), so the −15% is not an apples-to-apples margin.

Takeaway: scalloping costs $\approx 9\%$ drag at $< 1\%$ heat-flux change; peak G-load 16.83 sits *below* the 19.7 flight target.

Validation Progress — Convergence & DSMC/PINN

Convergence audit

(convergence_audit.csv)

Quantity	Value
C_d noise σ	0.0501
C_d R^2	0.8485
raw / denoised / PINN C_d	1.4625 / 1.4583 / 1.4632
PINN error vs DSMC	0.46%
$\dot{q}_{avg}$ R^2	0.9387
$\dot{q}_{max}$ R^2	0.7618

DSMC → PINN switch

- ▶ Hand-off at step 2200 → 3×10^8
- ▶ DSMC finals: $\dot{q}_{avg} = 56.59 \text{ W/cm}^2$, drag 45,410 N, $C_d = 1.4625$, $C_l = -0.556$, G 16.83, heat load 165.72 J/cm^2
- ▶ Kriging GP denoise bridges noise before PINN NS regression

Pipeline: SPARTA DSMC → Kriging GP denoise → PINN (Navier–Stokes) → BO optimization (≈ 2200 steps, 19,321 cells, 1000+ samples).

Optimization Progress — Bayesian Optimization

MoP replaced by Bayesian Optimization (commit 5bcb77), documented in `Optimization_Attempt_1.md` (21 Sep 2026).

Configuration

Item	Value
Algorithm	Bayesian Optimization
Surrogate	GP (Matern 5/2)
Acquisition	EI ($\xi = 0.01$)
Initial sampling	Latin Hypercube ($n = 20$)
BO iterations	50
Total evaluations	70 (20 LHD + 50 BO)
Cost function	C_d via Ada FFI

Design bounds

Parameter	Low	High
R_N [m]	0.5	3.0
r_{tor} [m]	0.02	0.25
half-cone [°]	30	80

Verified: trajectory

simulation runs step $0 \rightarrow 3 \times 10^8$ with optimized geometry.

Before vs After — Geometry Topology

BEFORE — default (IRVE-3 baseline)

Parameter	Value
R_N	1.5000 m
r_{tor}	0.1350 m
half-cone	60.00°
C_d	1.607278
SG heat flux	16.14 W/cm ²



AFTER — BO-optimized

Parameter	Value
R_N	1.1146 m (−25.7%)
r_{tor}	0.0539 m (−60.1%)
half-cone	44.58° (−25.7%)
C_d	1.455433 (−9.45%)
SG heat flux	18.72 W/cm ² (+16.0%)

Physics: smaller nose/torus radius cuts frontal area; narrower half-cone cuts wave drag.

Trade-off: sharper nose raises Sutton-Graves stagnation heating by 16% — must be priced against heat-shield mass in the next optimization pass.

Before vs After — Software Topology

BEFORE (deprecated 2026-08-21):

`main.py` → `StellarOrionEngineMach5Up.py` (Python orchestrator) → `HIAD_GeometryEngine.py` / Docker
SPARTA / visualizer / torch

AFTER (current):

`hiad_optimizer.py` → `ada_pinn_wrapper.py` (ctypes FFI) → **Ada/SPARK binary** `libstellarorion_pinn` →
SPARTA DSMC + PINN NS

Why: typed contracts and gnatprove-provable core (`prove.sh`, `gnatprove-*` logs, level-4 checks) replace a mutable Python engine; Python retains orchestration, BO, and plotting only.

Performance References — Flight & Literature Data

Reference	D [m]	$\dot{q}_{peak}$	Q [J/cm ²]	Notes
IRVE-3 flight	3.0	14.36 W/cm ²	195.06	19.7 g
LOFTID flight	6.0	39.27 W/cm ²	3520	9.66 g
Orion (ref.)	5.02	~ 1000 W/cm ²	—	$\beta \approx 400$
StellarOrion	3.0	56.59 W/cm ²	165.72	DSMC single-point

Rapisarda Table 4.10 vs IRVE-3 flight (trajectory-integrated, MCD v6.1):

- ▶ Fay-Riddell 13.8313 W/cm² (−3.69%, $R^2 = 0.9979$)
- ▶ Detra-Kemp-Riddell 14.0032 (−2.49%) · Chapman 13.9558 (−2.82%) · Van Driest 12.6375 (−12.0%)
- ▶ Sutton-Graves 15.2595 (+6.26%, $R^2 = 0.9603$)

Caveat: our DSMC value is a *fixed-condition snapshot* (alt 51.8 km, Mach 10.29, ISA, scalloped) — not directly comparable to trajectory-integrated flight numbers.

Performance References — Geometry Fidelity

- ▶ `REFERENCE_COMPARISON.md`: **all 18 geometry parameters** (5 primary + 13 derived/auxiliary) match Rapisarda Table 4.1 at **0.00%** delta each → **PASS**
- ▶ IRVE-3 baseline: 3.0 m, $R_n = 0.55$ m, 60° sphere-cone, 7 tori (6 in Rapisarda model), 281 kg, peak heat flux > 12 W/cm²
- ▶ Stale 0.55 m nose-radius references purged from Python (commit 3df86ad)
- ▶ Unified comparison (`unified_comparison_table.csv`): peak-G 16.83 vs 19.7 (−14.6%), heat load 165.7 vs 195.1 (−15.0%), C_d 1.4632 — PINN tracks Kriging/DSMC within $< 1\%$

Optimization Performance — Results

Metric	Default	Optimized	Change
C_d (cost)	1.607278	1.455433	− 9.45%
R_N [m]	1.5000	1.1146	−25.7%
r_{tor} [m]	0.1350	0.0539	−60.1%
half-cone [°]	60.00	44.58	−25.7%
SG heat flux [W/cm ²]	16.14	18.72	+16.0%

Design-space sampling (CCD/LHD phase, 15 probes):

best probe axial_r_tor_— $C_d = 1.542$; worst factorial+++ ($R_N=1.8$, $r_{tor}=0.18$, 65°) $C_d = 38.37$ —
response surface is strongly convex in r_{tor} and half-cone.

Convergence: 70 evaluations total (20 LHD space-filling + 50 EI-driven BO) → optimum at interior bounds, no constraint violation ($R_N \in [0.5, 3]$, $r_{tor} \in [0.02, 0.25]$, $\theta \in [30, 80]$).

Optimization Performance — Artefacts & Next Steps

Artefacts delivered this week

- ▶ `hiad_optimization_results.json` — default, optimized, improvement, 15 CCD samples, full BO config
 - ▶ `optimization_comparison.png` + `render_optimization_comparison.py` (new, untracked)
 - ▶ `hiad_optimized_overlay.mp4` — before/after geometry overlay animation
 - ▶ `Optimization_Attempt_1.md` — method, bounds, results, findings
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Next steps

- ▶ Multi-objective pass: C_d vs Sutton-Graves heating (currently -9.45% drag buys $+16\%$ heating)
- ▶ Re-run full 2200-step DSMC on optimized geometry for measured (not correlation-based) C_d and $\dot{q}$
- ▶ Commit working-tree churn in `_proc` and pin the comparison-render script

Week 13 Summary

Validation pipeline unblocked $\rightarrow$ converged;
first Bayesian optimization $\rightarrow C_d -9.45\%$.